

CASE STUDY | AI AUTOMATION

BizGrow AI Agent: Form-to-Email Workflow Built in n8n

A workflow that captures a business request through a form, sends it to an AI agent, and emails the AI-generated reply automatically, plus an honest review of what works and what to improve.

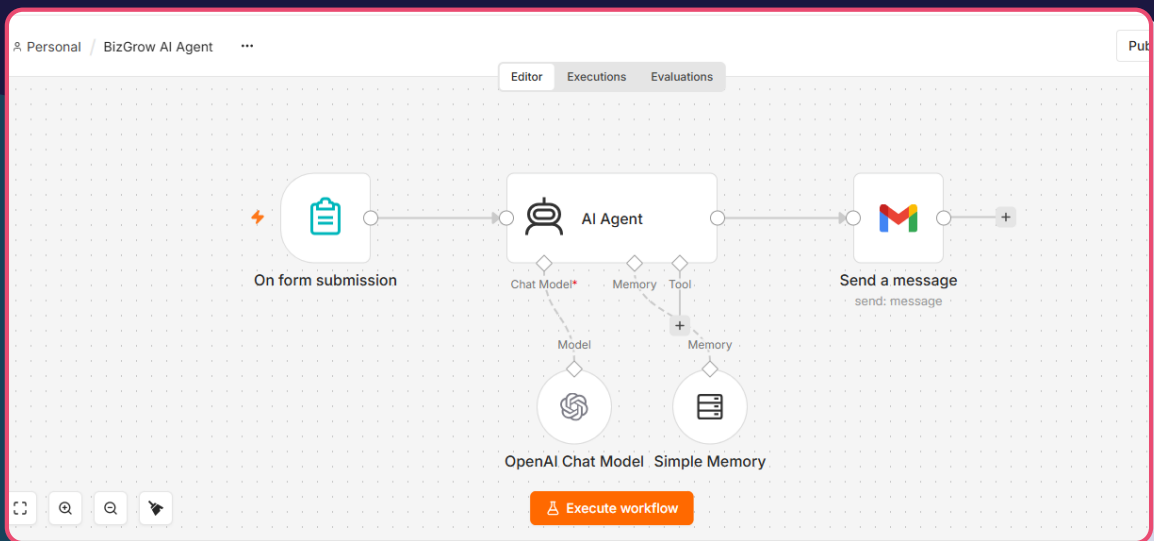
n8n

AI Agent

OpenAI

Simple Memory

Gmail



Salma Mazhar

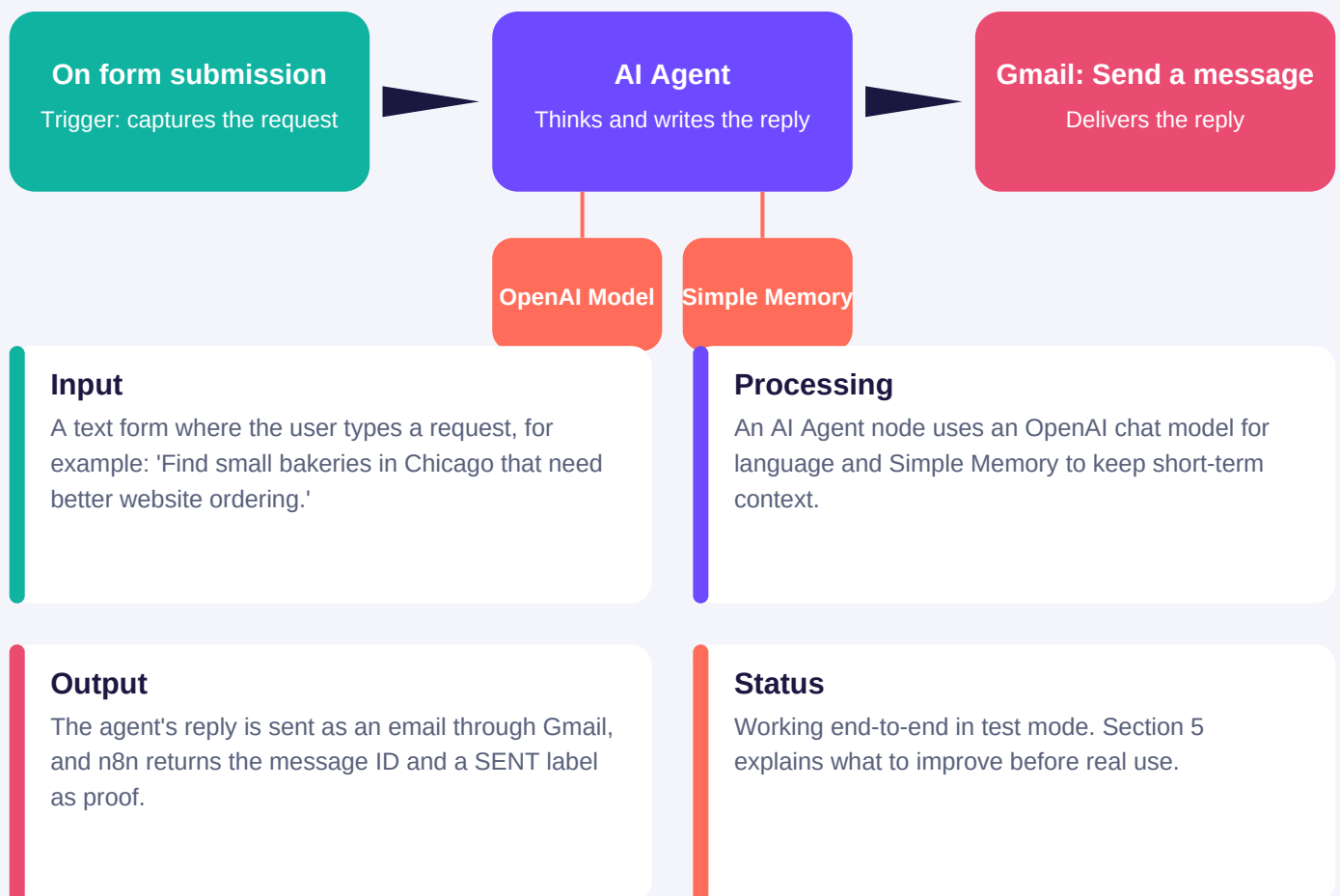
AI workflow automation | Product thinking

Project Overview

What the workflow does and why

Small businesses often want to reach out to local companies but do not have time to research each one. BizGrow AI Agent is a prototype of a 'lead finder' assistant: a person types a request in a form, an AI agent processes it, and the answer lands in an inbox without any manual work.

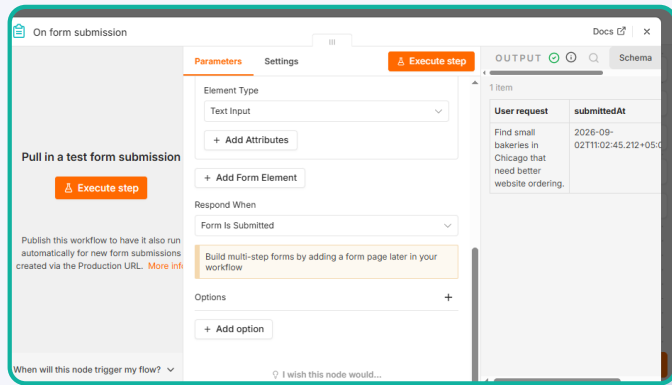
How it flows



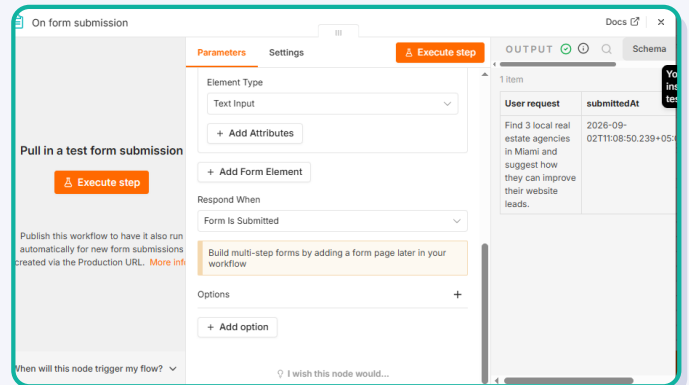
Step 1: The Form Trigger

Capturing the user's request

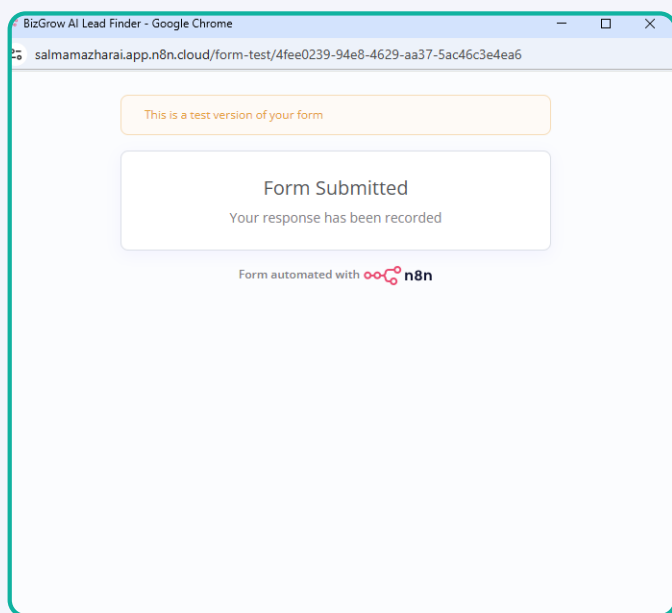
The 'On form submission' node uses a Text Input element so users can type a request freely. Each submission arrives with the request text and a submittedAt timestamp. Two different test requests were captured correctly.



Test 1: a bakery request in Chicago was captured with a timestamp.



Test 2: a Miami real-estate request was captured.



The live test form confirms: 'Your response has been recorded'.

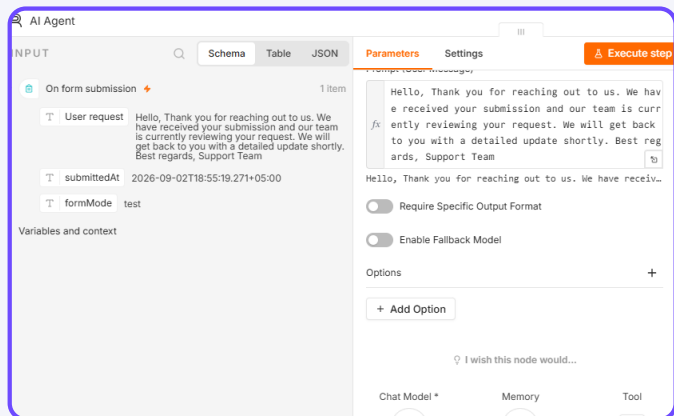
What this proves

The trigger works with different kinds of input, the form can be opened in a browser, and users get instant confirmation that their submission was received. This is the foundation the rest of the workflow depends on.

Step 2: The AI Agent

Where the thinking happens

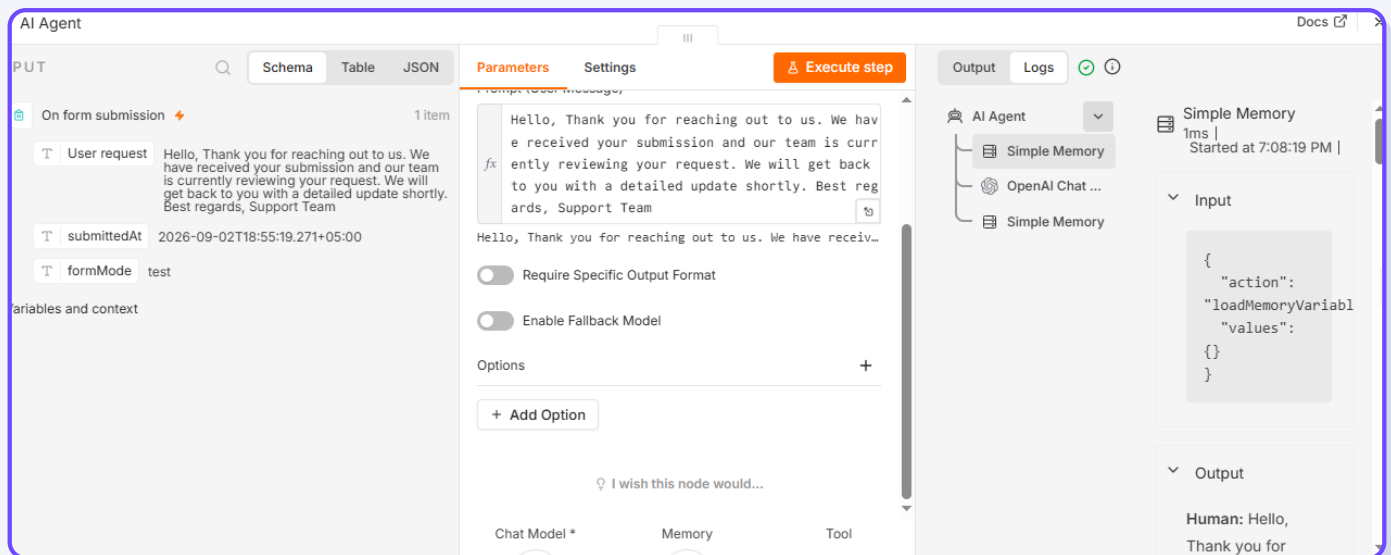
The AI Agent node receives the submission data (User request, submittedAt, formMode) and uses the connected OpenAI Chat Model to write a response. Simple Memory is switched on so the agent can keep conversation context.



Agent input data and the prompt that was sent to the model.

What the agent receives

Three fields arrive from the form: the request text, a timestamp and the form mode ('test'). The prompt field is where the user's request should be inserted for the model to act on.



Execution log: the agent loads memory variables, calls the OpenAI model, then saves to memory.

Step 3: Sending the Reply

Gmail 'Send a message' node

The last node takes the agent's output and sends it through Gmail. The output panel returns a message ID, a thread ID and the SENT label, which confirms delivery from n8n's side.

Send a message

INPUT

AI Agent

output: Do you want this rewritten for a specific channel/ton...
Subject (email): We've received your submission...
1) Short / concise
Thanks for contacting us. We've received your submission and are reviewing it. We'll get back to you shortly.
Support Team
2) Formal / professional
Hello (Name),
Thank you for contacting (Company). We have received your submission and our team is currently reviewing it. You can expect a detailed update within (X business days/hours). If this is urgent, please contact us at (phone/...

On form submission

Variables and context

Parameters

To: recipient@example.com

Subject: Form submission Inquiry

Email Type: Text

Message: Do you want this rewritten for a specific channel/ton...

Options: + Add option

OUTPUT

id	threadid	labels
1a0627575ca1cf49	1a0627575ca1cf49	SENT

Input: the agent's reply. Output: message ID, thread ID and 'SENT'. (Email address replaced for privacy.)

Confirmed

Gmail accepted the message and labelled it SENT. The full chain, form to agent to email, ran without manual steps.

Worth noticing

The email body begins with a question about rewriting the text. The next section explains why that happened.

Key takeaway

Connecting the three nodes is the easy part. The quality of the result now depends on what the agent is told to do and what it is allowed to use. That is the focus of the analysis on the next page.

Analysis and Findings

A product-minded review of the test run

What works

- + The trigger, agent and Gmail nodes are connected correctly and run in order.
- + Two different requests were captured, so the form is reliable.
- + OpenAI model and memory are wired into the agent.
- + Gmail returned SENT, so delivery works technically.

Gaps found in the screenshots

The agent is not answering the real request

High

In the test, the prompt held a ready-made 'Thank you for reaching out' message instead of the user's request. The agent treated it as text to polish, so the email offers rewrite options rather than leads.

No role instructions and no tools

High

The agent has no system prompt saying it is a lead finder, and the Tool slot is empty, so it cannot search the web for real businesses.

Recipient and subject are fixed

Med

The 'To' address and the subject 'Form submission Inquiry' are typed in manually. The form has no email field, so replies cannot reach the requester.

Memory is not separated per user

Med

Simple Memory without a session key can mix context between different submissions.

Test mode only

Low

The form shows 'test version' and formMode is 'test'. The workflow is not yet published for real users.

Improvement Roadmap

Turning the prototype into a useful product

1

Add an Email field to the form

So every reply can go to the person who asked.

2

Send the real request to the agent

Map the form's 'User request' field into the agent prompt.

3

Write a system prompt

Give the agent a clear role, format and limits (example below).

4

Connect a search tool

Let the agent look up real local businesses instead of guessing.

5

Make email dynamic

Use the requester's address and a subject built from their request.

6

Add session memory and logging

Separate memory per user and save each run to Google Sheets.

7

Publish and monitor

Switch from test to production URL and review the Executions tab.

Proposed system prompt (not yet built)

You are a lead research assistant for BizGrow, a small web agency. For each request, list up to 3 real local businesses, explain in one line why each could use a better website, and suggest one practical improvement. Write a short, friendly email in plain text. If you are unsure, say so instead of inventing details.

Lessons Learned

What this build taught me

A connected workflow is not a finished product

All nodes ran and Gmail returned SENT, yet the email was not useful. Success needs checking the content, not just the status.

The prompt is the product

The agent only knows what it is told. Clear role, input and output format decide the quality of every reply.

Design for the end user

Without an email field, the system cannot answer the person who asked. User needs should come before node settings.

Test, read, then iterate

Reading the execution log and each output showed exactly where the chain went off track, which made the fixes easy to list.

Need an AI workflow for your business?

I build practical automations with n8n, Zapier, Google Workspace and AI agents. I also review existing workflows and turn them into reliable, user-friendly systems.

Salma Mazhar

Message me to discuss your project.